PG No. 19 C_{3v} $3m$ (3m1 setting) [trigonal] (internal axial dipole)

* Harmonics for rank 0

$\vec{\mathbb{G}}_0^{(1,1)}[g](A_2)$

** symmetry

1

** expression

$$\frac{\sqrt{3}G_x x}{3} + \frac{\sqrt{3}G_y y}{3} + \frac{\sqrt{3}G_z z}{3}$$

* Harmonics for rank 1

$\vec{\mathbb{Q}}_1^{(1,0)}[g](A_1)$

** symmetry

z

** expression

$$\frac{\sqrt{2}G_x y}{2} - \frac{\sqrt{2}G_y x}{2}$$

$\vec{\mathbb{G}}_1^{(1,-1)}[g](A_2)$

** symmetry

z

** expression

G_z

$\vec{\mathbb{G}}_1^{(1,1)}[g](A_2)$

** symmetry

z

** expression

$$\frac{3\sqrt{10}G_x xz}{10} + \frac{3\sqrt{10}G_y yz}{10} - \frac{\sqrt{10}G_z\left(x^2+y^2-2z^2\right)}{10}$$

$\vec{\mathbb{Q}}_{1,1}^{(1,0)}[g](E)$, $\vec{\mathbb{Q}}_{1,2}^{(1,0)}[g](E)$

** symmetry

x

y

** expression

$$\frac{\sqrt{2}G_y z}{2} - \frac{\sqrt{2}G_z y}{2}$$

$$-\frac{\sqrt{2}G_x z}{2} + \frac{\sqrt{2}G_z x}{2}$$

$\vec{\mathbb{G}}_{1,1}^{(1,-1)}[g](E)$, $\vec{\mathbb{G}}_{1,2}^{(1,-1)}[g](E)$

** symmetry

$-y$

x

** expression

$-G_y$

G_x

$\vec{\mathbb{G}}_{1,1}^{(1,1)}[g](E)$, $\vec{\mathbb{G}}_{1,2}^{(1,1)}[g](E)$

** symmetry

$-y$

x

** expression

$$-\frac{3\sqrt{10}G_x xy}{10} + \frac{\sqrt{10}G_y\left(x^2-2y^2+z^2\right)}{10} - \frac{3\sqrt{10}G_z yz}{10}$$

$$\frac{\sqrt{10}G_x\left(2x^2-y^2-z^2\right)}{10}+\frac{3\sqrt{10}G_yxy}{10}+\frac{3\sqrt{10}G_zxz}{10}$$

* Harmonics for rank 2

$\vec{\mathbb{Q}}_2^{(1,0)}[g](A_1)$

** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$\frac{\sqrt{6}G_xyz}{2}-\frac{\sqrt{6}G_yxz}{2}$$

$\vec{\mathbb{G}}_2^{(1,-1)}[g](A_2)$

** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$-\frac{\sqrt{6}G_xx}{6}-\frac{\sqrt{6}G_yy}{6}+\frac{\sqrt{6}G_zz}{3}$$

$\vec{\mathbb{G}}_2^{(1,1)}[g](A_2)$

** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$-\frac{\sqrt{21}G_xx\left(x^2+y^2-4z^2\right)}{14}-\frac{\sqrt{21}G_yy\left(x^2+y^2-4z^2\right)}{14}-\frac{\sqrt{21}G_zz\left(3x^2+3y^2-2z^2\right)}{14}$$

$\vec{\mathbb{Q}}_{2,1}^{(1,0)}[g](E,1)$, $\vec{\mathbb{Q}}_{2,2}^{(1,0)}[g](E,1)$

** symmetry

$$\sqrt{3}xz$$

$$\sqrt{3}yz$$

** expression

$$\frac{\sqrt{2}G_xxy}{2}-\frac{\sqrt{2}G_y\left(x-z\right)\left(x+z\right)}{2}-\frac{\sqrt{2}G_zyz}{2}$$

$$\frac{\sqrt{2}G_x\left(y-z\right)\left(y+z\right)}{2}-\frac{\sqrt{2}G_yxy}{2}+\frac{\sqrt{2}G_zxz}{2}$$

$\vec{\mathbb{Q}}_{2,1}^{(1,0)}[g](E,2)$, $\vec{\mathbb{Q}}_{2,2}^{(1,0)}[g](E,2)$

** symmetry

$$\sqrt{3}xy$$

$$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$$

** expression

$$-\frac{\sqrt{2}G_xxz}{2}+\frac{\sqrt{2}G_yyz}{2}+\frac{\sqrt{2}G_z\left(x-y\right)\left(x+y\right)}{2}$$

$$\frac{\sqrt{2}G_xyz}{2}+\frac{\sqrt{2}G_yxz}{2}-\sqrt{2}G_zxy$$

$\vec{\mathbb{G}}_{2,1}^{(1,-1)}[g](E,1)$, $\vec{\mathbb{G}}_{2,2}^{(1,-1)}[g](E,1)$

** symmetry

$$\sqrt{3}yz$$

$$-\sqrt{3}xz$$

** expression

$$\frac{\sqrt{2}G_yz}{2}+\frac{\sqrt{2}G_zy}{2}$$

$$-\frac{\sqrt{2}G_x z}{2} - \frac{\sqrt{2}G_z x}{2}$$

$\vec{\mathbb{G}}_{2,1}^{(1,-1)}[g](E,2)$, $\vec{\mathbb{G}}_{2,2}^{(1,-1)}[g](E,2)$

** symmetry

$$\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$$

$$-\sqrt{3}xy$$

** expression

$$\frac{\sqrt{2}G_x x}{2} - \frac{\sqrt{2}G_y y}{2}$$

$$-\frac{\sqrt{2}G_x y}{2} - \frac{\sqrt{2}G_y x}{2}$$

$\vec{\mathbb{G}}_{2,1}^{(1,1)}[g](E,1)$, $\vec{\mathbb{G}}_{2,2}^{(1,1)}[g](E,1)$

** symmetry

$$\sqrt{3}yz$$

$$-\sqrt{3}xz$$

** expression

$$\frac{5\sqrt{7}G_x xyz}{7} - \frac{\sqrt{7}G_y z\left(x^2-4y^2+z^2\right)}{7} - \frac{\sqrt{7}G_z y\left(x^2+y^2-4z^2\right)}{7}$$

$$-\frac{\sqrt{7}G_x z\left(4x^2-y^2-z^2\right)}{7} - \frac{5\sqrt{7}G_y xyz}{7} + \frac{\sqrt{7}G_z x\left(x^2+y^2-4z^2\right)}{7}$$

$\vec{\mathbb{G}}_{2,1}^{(1,1)}[g](E,2)$, $\vec{\mathbb{G}}_{2,2}^{(1,1)}[g](E,2)$

** symmetry

$$\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$$

$$-\sqrt{3}xy$$

** expression

$$\frac{\sqrt{7}G_x x\left(3x^2-7y^2-2z^2\right)}{14} + \frac{\sqrt{7}G_y y\left(7x^2-3y^2+2z^2\right)}{14} + \frac{5\sqrt{7}G_z z\left(x-y\right)\left(x+y\right)}{14}$$

$$-\frac{\sqrt{7}G_x y\left(4x^2-y^2-z^2\right)}{7} + \frac{\sqrt{7}G_y x\left(x^2-4y^2+z^2\right)}{7} - \frac{5\sqrt{7}G_z xyz}{7}$$

* Harmonics for rank 3

$\vec{\mathbb{Q}}_3^{(1,0)}[g](A_1,1)$

** symmetry

$$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$$

** expression

$$-\frac{\sqrt{3}G_x y\left(x^2+y^2-4z^2\right)}{4} + \frac{\sqrt{3}G_y x\left(x^2+y^2-4z^2\right)}{4}$$

$\vec{\mathbb{Q}}_3^{(1,0)}[g](A_1,2)$

** symmetry

$$\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$$

** expression

$$-\frac{\sqrt{30}G_x z\left(x-y\right)\left(x+y\right)}{8} + \frac{\sqrt{30}G_y xyz}{4} + \frac{\sqrt{30}G_z x\left(x^2-3y^2\right)}{8}$$

$\vec{\mathbb{G}}_3^{(1,-1)}[g](A_1)$

** symmetry

$$\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$$

** expression

$$\frac{\sqrt{6}G_x\left(x-y\right)\left(x+y\right)}{4} - \frac{\sqrt{6}G_yxy}{2}$$

$\vec{\mathbb{G}}_3^{(1,1)}[g](A_1)$
** symmetry

$$\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$$

** expression

$$\frac{\sqrt{10}G_x\left(4x^4-21x^2y^2-3x^2z^2+3y^4+3y^2z^2\right)}{24} + \frac{\sqrt{10}G_yxy\left(13x^2-15y^2+6z^2\right)}{24} + \frac{7\sqrt{10}G_zxz\left(x^2-3y^2\right)}{24}$$

$\vec{\mathbb{Q}}_3^{(1,0)}[g](A_2)$
** symmetry

$$\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$$

** expression

$$\frac{\sqrt{30}G_xxyz}{4} + \frac{\sqrt{30}G_yz\left(x-y\right)\left(x+y\right)}{8} - \frac{\sqrt{30}G_zy\left(3x^2-y^2\right)}{8}$$

$\vec{\mathbb{G}}_3^{(1,-1)}[g](A_2,1)$
** symmetry

$$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$$

** expression

$$-\frac{\sqrt{15}G_xxz}{5} - \frac{\sqrt{15}G_yyz}{5} - \frac{\sqrt{15}G_z\left(x^2+y^2-2z^2\right)}{10}$$

$\vec{\mathbb{G}}_3^{(1,-1)}[g](A_2,2)$
** symmetry

$$\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$$

** expression

$$\frac{\sqrt{6}G_xxy}{2} + \frac{\sqrt{6}G_y\left(x-y\right)\left(x+y\right)}{4}$$

$\vec{\mathbb{G}}_3^{(1,1)}[g](A_2,1)$
** symmetry

$$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$$

** expression

$$-\frac{5G_xxz\left(3x^2+3y^2-4z^2\right)}{12} - \frac{5G_yyz\left(3x^2+3y^2-4z^2\right)}{12} + \frac{G_z\left(3x^4+6x^2y^2-24x^2z^2+3y^4-24y^2z^2+8z^4\right)}{12}$$

$\vec{\mathbb{G}}_3^{(1,1)}[g](A_2,2)$
** symmetry

$$\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$$

** expression

$$\frac{\sqrt{10}G_xxy\left(15x^2-13y^2-6z^2\right)}{24} - \frac{\sqrt{10}G_y\left(3x^4-21x^2y^2+3x^2z^2+4y^4-3y^2z^2\right)}{24} + \frac{7\sqrt{10}G_zyz\left(3x^2-y^2\right)}{24}$$

$\vec{\mathbb{Q}}_{3,1}^{(1,0)}[g](E,1)$, $\vec{\mathbb{Q}}_{3,2}^{(1,0)}[g](E,1)$
** symmetry

$$-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$$

$$-\frac{\sqrt{6}x^2y}{4} - \frac{\sqrt{6}y^3}{4} + \sqrt{6}yz^2$$

** expression

$$\frac{5\sqrt{2}G_x xyz}{4} - \frac{\sqrt{2}G_y z\left(11x^2+y^2-4z^2\right)}{8} + \frac{\sqrt{2}G_z y\left(x^2+y^2-4z^2\right)}{8}$$

$$\frac{\sqrt{2}G_x z\left(x^2+11y^2-4z^2\right)}{8} - \frac{5\sqrt{2}G_y xyz}{4} - \frac{\sqrt{2}G_z x\left(x^2+y^2-4z^2\right)}{8}$$

$\vec{\mathbb{Q}}_{3,1}^{(1,0)}[g](E,2)$, $\vec{\mathbb{Q}}_{3,2}^{(1,0)}[g](E,2)$

** symmetry

$$\sqrt{15}xyz$$

$$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{\sqrt{5}G_x x\left(y-z\right)\left(y+z\right)}{2} - \frac{\sqrt{5}G_y y\left(x-z\right)\left(x+z\right)}{2} + \frac{\sqrt{5}G_z z\left(x-y\right)\left(x+y\right)}{2}$$

$$\frac{\sqrt{5}G_x y\left(x^2-y^2+2z^2\right)}{4} - \frac{\sqrt{5}G_y x\left(x^2-y^2-2z^2\right)}{4} - \sqrt{5}G_z xyz$$

$\vec{\mathbb{G}}_{3,1}^{(1,-1)}[g](E,1)$, $\vec{\mathbb{G}}_{3,2}^{(1,-1)}[g](E,1)$

** symmetry

$$\frac{\sqrt{6}x^2y}{4} + \frac{\sqrt{6}y^3}{4} - \sqrt{6}yz^2$$

$$-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$$

** expression

$$\frac{\sqrt{10}G_x xy}{10} + \frac{\sqrt{10}G_y\left(x^2+3y^2-4z^2\right)}{20} - \frac{2\sqrt{10}G_z yz}{5}$$

$$-\frac{\sqrt{10}G_x\left(3x^2+y^2-4z^2\right)}{20} - \frac{\sqrt{10}G_y xy}{10} + \frac{2\sqrt{10}G_z xz}{5}$$

$\vec{\mathbb{G}}_{3,1}^{(1,-1)}[g](E,2)$, $\vec{\mathbb{G}}_{3,2}^{(1,-1)}[g](E,2)$

** symmetry

$$-\frac{\sqrt{15}x^2z}{2} + \frac{\sqrt{15}y^2z}{2}$$

$$\sqrt{15}xyz$$

** expression

$$-G_x xz + G_y yz - \frac{G_z\left(x-y\right)\left(x+y\right)}{2}$$

$$G_x yz + G_y xz + G_z xy$$

$\vec{\mathbb{G}}_{3,1}^{(1,1)}[g](E,1)$, $\vec{\mathbb{G}}_{3,2}^{(1,1)}[g](E,1)$

** symmetry

$$\frac{\sqrt{6}x^2y}{4} + \frac{\sqrt{6}y^3}{4} - \sqrt{6}yz^2$$

$$-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$$

** expression

$$\frac{5\sqrt{6}G_x xy\left(x^2+y^2-6z^2\right)}{24} - \frac{\sqrt{6}G_y\left(x^4-3x^2y^2-3x^2z^2-4y^4+27y^2z^2-4z^4\right)}{24} + \frac{5\sqrt{6}G_z yz\left(3x^2+3y^2-4z^2\right)}{24}$$

$$-\frac{\sqrt{6}G_x\left(4x^4+3x^2y^2-27x^2z^2-y^4+3y^2z^2+4z^4\right)}{24} - \frac{5\sqrt{6}G_y xy\left(x^2+y^2-6z^2\right)}{24} - \frac{5\sqrt{6}G_z xz\left(3x^2+3y^2-4z^2\right)}{24}$$

$\vec{\mathbb{G}}_{3,1}^{(1,1)}[g](E,2)$, $\vec{\mathbb{G}}_{3,2}^{(1,1)}[g](E,2)$

** symmetry

$$-\frac{\sqrt{15}x^2z}{2}+\frac{\sqrt{15}y^2z}{2}$$

$$\sqrt{15}xyz$$

** expression

$$-\frac{\sqrt{15}G_xxz\left(5x^2-9y^2-2z^2\right)}{12}-\frac{\sqrt{15}G_yyz\left(9x^2-5y^2+2z^2\right)}{12}+\frac{\sqrt{15}G_z\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{12}$$

$$\frac{\sqrt{15}G_xyz\left(6x^2-y^2-z^2\right)}{6}-\frac{\sqrt{15}G_yxz\left(x^2-6y^2+z^2\right)}{6}-\frac{\sqrt{15}G_zxy\left(x^2+y^2-6z^2\right)}{6}$$